# 03/07/2022

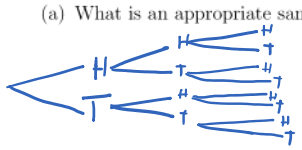
Monday, March 7, 2022 11:27 AM

Math_168_Exam-II_Review

Mr. Orchard's Math 141 WIR Test 2 Review Week 7

1. An experiment consists of tossing three coins and observing the results.

   (a) What is an appropriate sample space for this experiment?




$\{HHH, \ldots, TTT\}$

   (b) Write the event of getting at least two heads in roster notation.

$\cdot E_{H\geq 2} = \{HHT, HTH, THH, HHH\}$

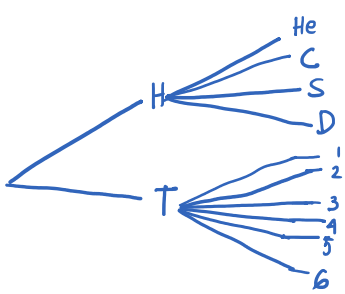
   (c) Are the events of getting at least two heads and getting exactly one tails mutually exclusive? Why or why not?

$\cdot E_{T=1} = \{THH, HTH, HHT\}.$

   (d) How many events does the sample space have?

$2^8 = 256$

2. An experiment consists of tossing a coin and observing the result. If the result is heads, we draw a card from a standard well-shuffled 52 card deck and record the suit. if the result is tails, we roll a fair six sided die and record the result. What is an appropriate sample space for this experiment?




$\{H\,He, HC, HS, HD, T1, T2, T3, T4, T5, T6\}$

3. A bag of Bertie Bott's Every Flavor Beans contains 12 chalk dust flavored beans, 7 cat toe flavored beans, and 7 raspberry flavored beans. A sample of 12 of them is to be picked from the bag.

   (a) How many samples contain at least one chalk dust flavored bean?

   (b) How many samples contain exactly 4 chalk dust and exactly 6 cat toe flavored beans?

   (c) How many samples contain exactly 4 chalk dust or exactly 6 cat toe flavored beans?

4. Three families, each with four people in them, are sitting in a movie theater row.

   (a) In how many ways can the twelve people be seated in the row if there are 12 seats in the row?

$N! = 1 \cdot 2 \cdot 3 \cdots N$

$3! = 1 \cdot 2 \cdot 3 = 6$

$12!$

12 11 10 ... 1

   (b) In how many ways can the twelve people be seated in the row if there are 14 seats in the row, the families must sit together, and there must be an empty seat between each of the families?

$F_1$ $F_2$ $F_3$

12 3 2 1 8 3 2 1 4 3 2 1 $= \boxed{(3!)^3 \times 12 \times 8 \times 4} = \boxed{82{,}944}$

The first chair can be filled in 12 different ways.

$\boxed{(4!)^3 \times 3!} = \boxed{82{,}944}$

3 2 1

5. Mallorie has 4 red pens, 3 blue pens, 7 black pens, an orange pen, and a green pen. How many ways can she arrange her pens on her desk in a distinguishable arrangement?

6. (a) In how many ways can a 2 card hand be drawn from a standard 52 card deck?

$$C(52,2) = \frac{52!}{50! \times 2!} = 1326$$

(b) In how many ways can a 2 cards of the same suit be drawn from a standard 52 card deck?

$$C(4,1) \times C(13,2) = \boxed{\frac{4!}{3! \times 1!} \times \frac{13!}{11! \times 2!}} = \boxed{312}$$

(c) In how many ways can a 2 card hand be drawn if they are not both the same suit?

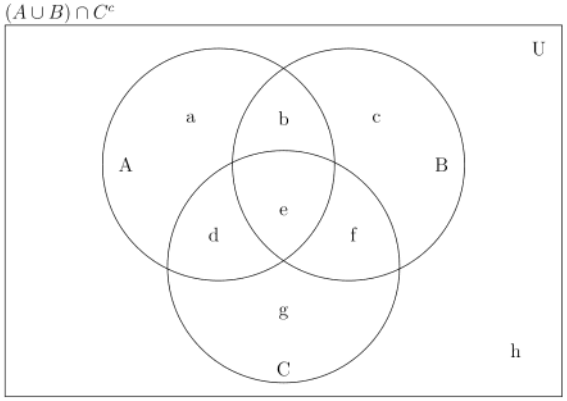
$$1326 - 312$$

(d) In how many ways can a pair be drawn in a two card hand?

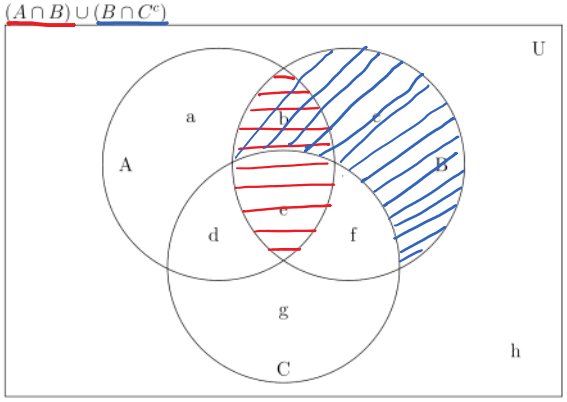
$$C(13,1) \times C(4,2) = \boxed{78}$$

7. Shade the Venn diagrams to indicate the given sets.

(a) $(A \cup B) \cap C^c$




(b) $(A \cap B) \cup (B \cap C^c)$

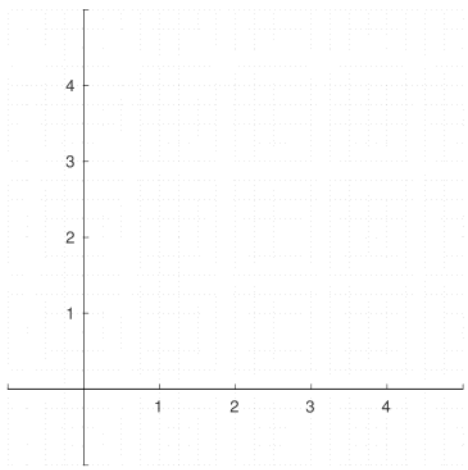


8. Use the below information to figure out how many people drink only coffee.
   A survey of 181 people were asked if they drink hot chocolate, tea, or coffee.
   6 people drink all three.
   38 people drink tea and hot chocolate.
   40 people drink hot chocolate and coffee, but not tea.
   65 people drink tea but not coffee.
   88 people drink hot chocolate.
   94 people do not drink coffee.
   78 people drink exactly one of these.

9. Consider the linear programming problem below:

$$\begin{aligned} \text{Maximize:} \quad & P = 2x + 3y \\ \text{subject to:} \quad & x + y \leq 3 \\ & 2x + y \leq 5 \\ & 2x - y \geq -1 \\ & x \geq 0 \\ & y \geq 0 \end{aligned}$$

(a) Determine graphically the solution set for the inequalities.




(b) Is the solution set bounded or unbounded? Justify your answer.

(c) Using the method of corners, find the $x$- and $y$-coordinates that solve the linear programming problem.